

# Education Subsidy as Insurance for Fertility Choices\*

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*Preliminary and incomplete*

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## Abstract

Macroeconomic challenges associated with low fertility have driven many countries to implement pro-natal policies. I study the consequences of a novel policy recently implemented in Japan, which subsidizes college students, given that college education comprises more than half of the total education expenses incurred by average parents to raise a child. I construct a lifecycle Aiyagari model with college enrollment and fertility choices, where altruistic parents make asset transfers to their children to support their college education. The model is calibrated using Japanese household panel data and successfully replicates several measures of fertility elasticities observed or estimated empirically. I find that an income-tested college subsidy acts as partial insurance for fertility choices; having children is not only expensive but also risky under income uncertainty because, with more children, a bad income shock is more likely to result in a shortfall in financing their children's education, leading to utility losses for altruistic parents. The income-tested subsidy partially insures against such costly states. The resulting fertility responses amplify the macroeconomic effects of the college subsidy—on inequality reduction, welfare improvement, and increasing income levels—through the associated general equilibrium effects and intergenerational linkages.

Keywords: Fertility, incomplete market, overlapping generations, college subsidy

JEL codes: C68, I28, J13, J24

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